Supplementary Information: Explainable Machine Learning for Predicting Mathematics Literacy and Low-Performing Students: Evidence from PISA 2022

Zhiren Xiao[1*]

[1*]Guangdong University of Finance, China.

Corresponding author(s). E-mail(s): 241734106@m.gduf.edu.cn;

Supplementary Tables

Table 1 ALE-based feature importance and comparison with SHAP-based rankings for the best classification model (tuned XGBoost). First-order ALE was computed on a 10,000-row explanation sample with 20 bins.

Metric	Value
Spearman's ρ (ALE vs. SHAP importance)	0.76
Kendall's τ (ALE vs. SHAP importance)	0.58
p-value (ALE–SHAP rank correlation)	< 0.001
Top-ranked predictors (by ALE)	
Rank 1	Home possessions (HOMEPOS)
Rank 2	Mathematics self-efficacy (MATHEFF)
Rank 3	Grade
Notable ranking divergences (SHAP vs. ALE)	
ICT self-efficacy (ICTEFFIC)	Higher ALE rank than SHAP rank
ICT resources (ICTRES)	Lower ALE rank than SHAP rank

The top three predictors by ALE were identical to those by permutation importance, providing cross-method validation. Divergences for ICT variables likely reflect differing treatment of feature correlations: SHAP assumes independence, inflating importance of features with broader marginal distributions, while ALE conditions on local intervals.

Table 2 Formal algorithmic fairness metrics for the best classification model (tuned XGBoost). Equalized Odds Difference, Demographic Parity Difference, and ABROCA (Absolute Between-ROC Area) are reported across gender, immigrant background, and ESCS quartile groups.

Group	Equal. Odds Diff. (TPR)	Dem. Parity Diff.	ABROCA
Gender (Female vs. Male)			
Female vs. Male	< 0.01	0.02	0.008
Immigrant background (Native vs. Non-native)			
Native vs. Non-native	0.04	0.03	0.015
ESCS quartile (Q4–privileged vs. Q1–unprivileged)			
Q4 vs. Q1	0.07	–	0.023

ABROCA values below 0.01 are generally considered negligible, 0.01–0.03 modest but noteworthy, and above 0.03 potentially actionable (**??**). By these benchmarks: gender disparities are negligible, immigrant-background disparities are modest, and SES-based disparities approach the actionable threshold. Intersectional analysis additionally showed that the lowest-ESCS, non-native-immigrant subgroup had AUC = 0.831 and F1 = 0.712, compared with AUC = 0.903 and F1 = 0.824 for the highest-ESCS, native subgroup.

Table 3 Structured comparison of the present study with published PISA-ML studies from 2024–2025. Performance metrics, sample characteristics, and methodological features are compared.

Study	Sample	Countries	AUC	R^2/RMSE	XAI	Fairness
This study	613,744	80 (global)	0.903	0.681/54.10	Multi-method	Formal
?	∼14,000	1 (Australia)	–	0.42/–	–	–
?	∼30,800	1 (Spain)	0.977	–	SHAP + UMAP	–
?	34,968	6 (East Asia)	–	–	SHAP	–
?	∼613,000	80 (global)	–	–	None	None
?	∼500,000	79 (global)	–	–	SHAP	Subgroup
?	4,552	1 (USA)	–	–	SHAP + ALE	–
?	∼600,000	76 (global)	0.76–0.93	–	SHAP	–

Dashes indicate metrics or features not reported in the published study. Comparisons are offered as contextual benchmarks rather than competitive claims. The wide AUC range for **?** reflects per-country model performance.

Table 4 Explanation stability across ESCS quintile subgroups. Pairwise Spearman's ρ rank correlation of SHAP-based feature importance rankings between ESCS quintiles. All correlations are significant at $p < 0.001$ except Q1–Q5 ($p = 3 \times 10^{-6}$).

Subgroup pair	Spearman's ρ	p-value
ESCS quintile comparisons		
Q1 (lowest) vs. Q2	0.897	< 0.001
Q1 (lowest) vs. Q3	0.851	< 0.001
Q1 (lowest) vs. Q4	0.811	< 0.001
Q1 (lowest) vs. Q5 (highest)	0.715	3×10^{-6}
Q2 vs. Q3	0.987	< 0.001
Q2 vs. Q4	0.962	< 0.001
Q2 vs. Q5 (highest)	0.847	< 0.001
Q3 vs. Q4	0.984	< 0.001
Q3 vs. Q5 (highest)	0.860	< 0.001
Q4 vs. Q5 (highest)	0.873	< 0.001

Feature importance rankings were computed using SHAP on the best classification model (tuned XGBoost) with 33 shared features. Higher ESCS quintiles indicate greater socioeconomic advantage. The lowest cross-group consistency (Q1 vs. Q5, $\rho = 0.72$) suggests modestly divergent feature importance patterns between the most and least advantaged students.

Table 5 Multi-method XAI comparison: rank correlation between feature importance rankings from SHAP, permutation importance, and LIME for the best classification model (tuned XGBoost).

Comparison	Spearman's ρ	Kendall's τ	p-value
SHAP vs. Permutation importance	0.831	0.663	< 0.001
SHAP vs. LIME	0.004	-0.004	0.982
Permutation importance vs. LIME	0.024	-0.008	0.896

SHAP and permutation importance showed high inter-method consistency, with the top two predictors (HOMEPOS, MATHEFF) identically ranked. LIME-based rankings diverged substantially, likely reflecting LIME's reliance on local linear approximations.

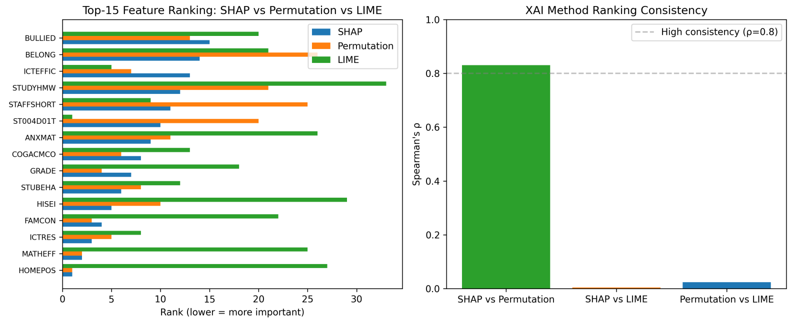

Fig. 1 Multi-method XAI comparison: feature importance rankings from SHAP, permutation importance, and LIME for the best classification model.

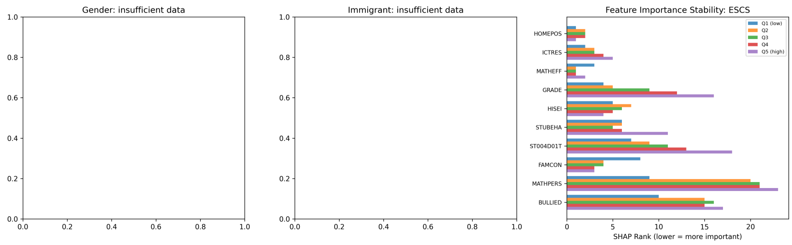

Fig. 2 Explanation stability across ESCS quintile subgroups: SHAP-based feature importance ranking consistency.

Supplementary Figures

SHAP Dependence Plots

The following figures show SHAP dependence plots for ICT interaction pairs, computed on a 5,000-row explanation sample using the best classification model (tuned XGBoost).

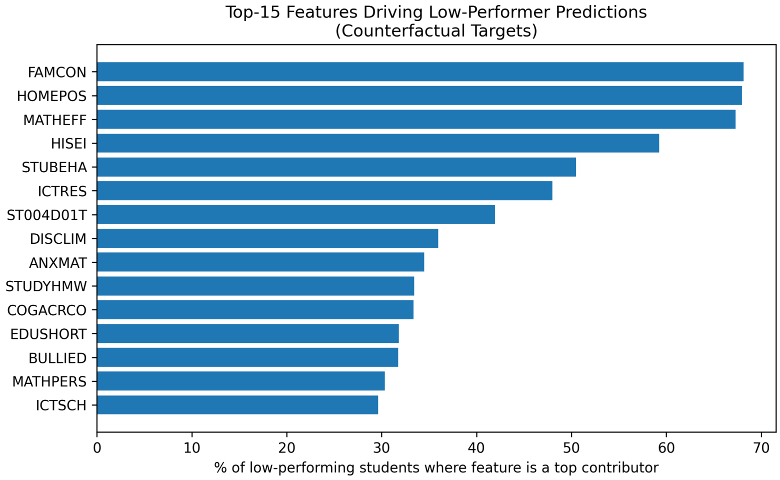

Fig. 3 Counterfactual feature importance: mean SHAP contribution for features appearing in counterfactual explanations.

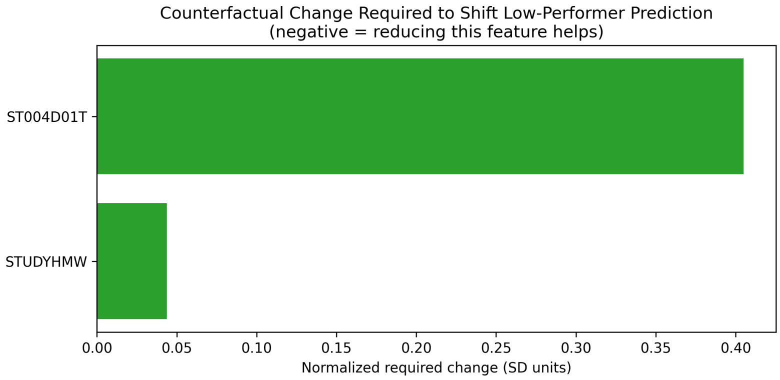

Fig. 4 Counterfactual magnitude analysis: required feature change to flip predictions.

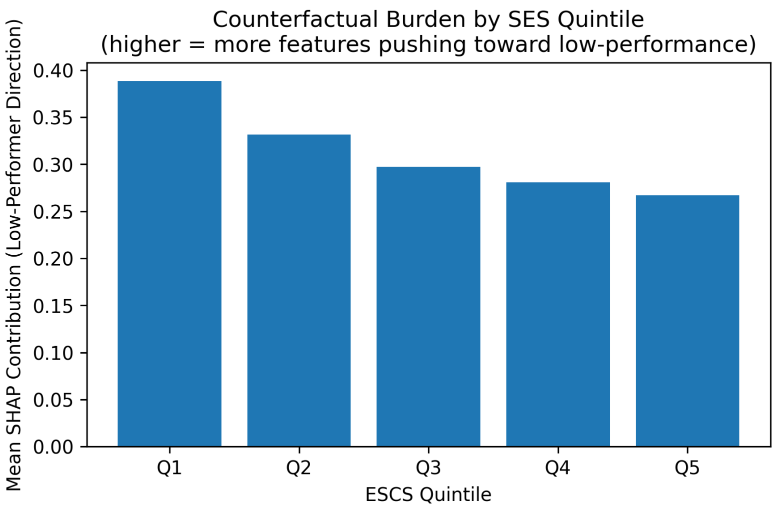

Fig. 5 Counterfactual reachability by ESCS quartile: the "burden" of required counterfactual change varies systematically by socioeconomic status.

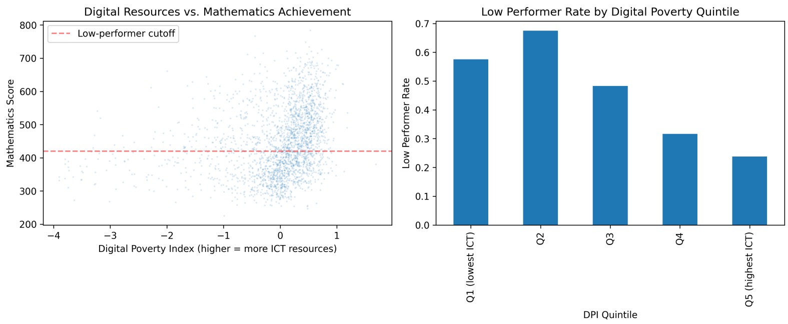

Fig. 6 Digital Poverty Index by ESCS quintile: composite measure of ICT access disadvantage.

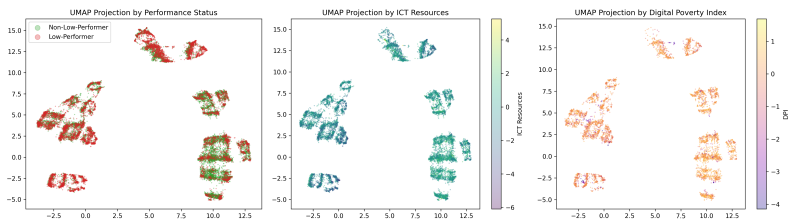

Fig. 7 UMAP projections of student profiles based on the 33 main-model features, colored by mathematics performance quintile and country income group.

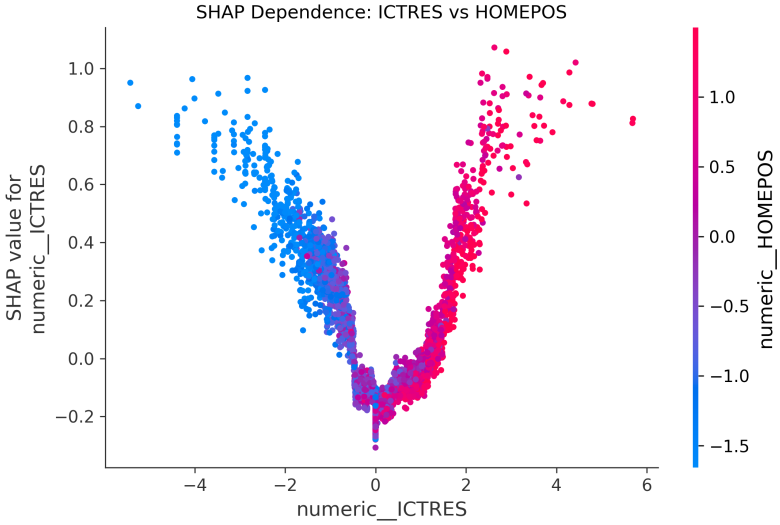

Fig. 8 SHAP dependence: ICT resources (ICTRES) vs. home possessions (HOMEPOS).

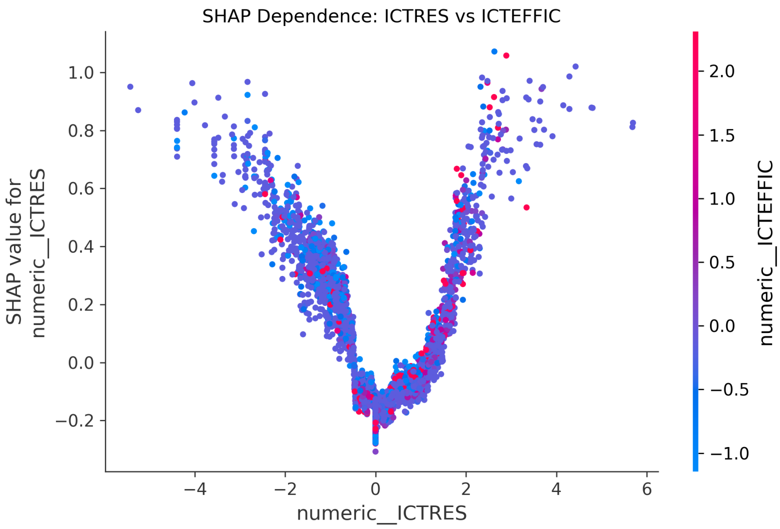

Fig. 9 SHAP dependence: ICT resources (ICTRES) vs. ICT self-efficacy (ICTEFFIC).

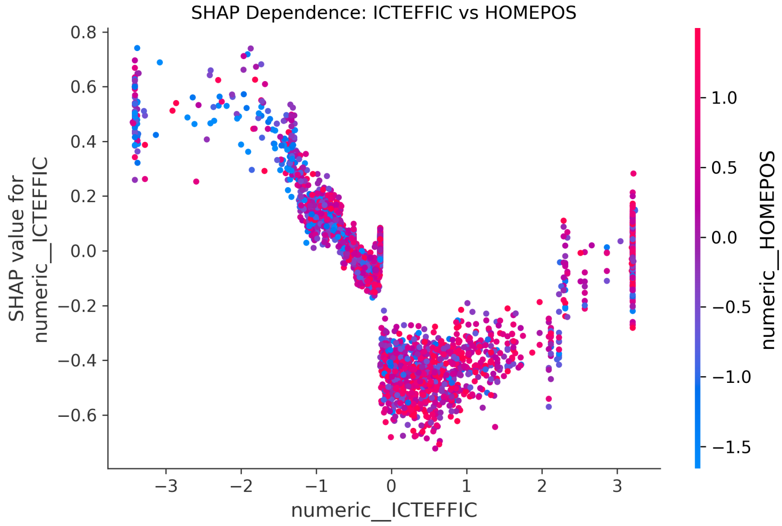

Fig. 10 SHAP dependence: ICT self-efficacy (ICTEFFIC) vs. home possessions (HOMEPOS).

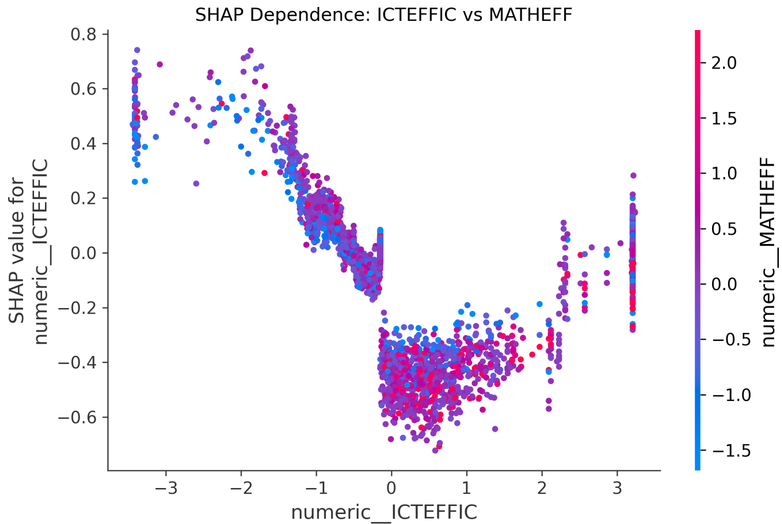

Fig. 11 SHAP dependence: ICT self-efficacy (ICTEFFIC) vs. mathematics self-efficacy (MATH-EFF).

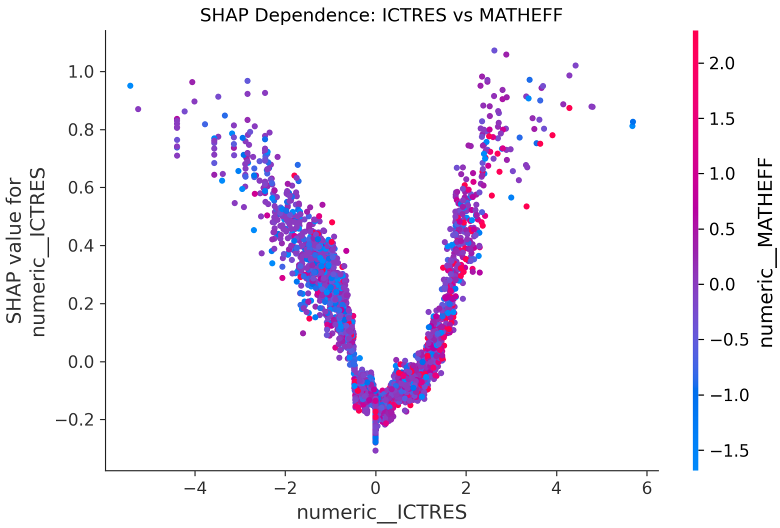

Fig. 12 SHAP dependence: ICT resources (ICTRES) vs. mathematics self-efficacy (MATHEFF).

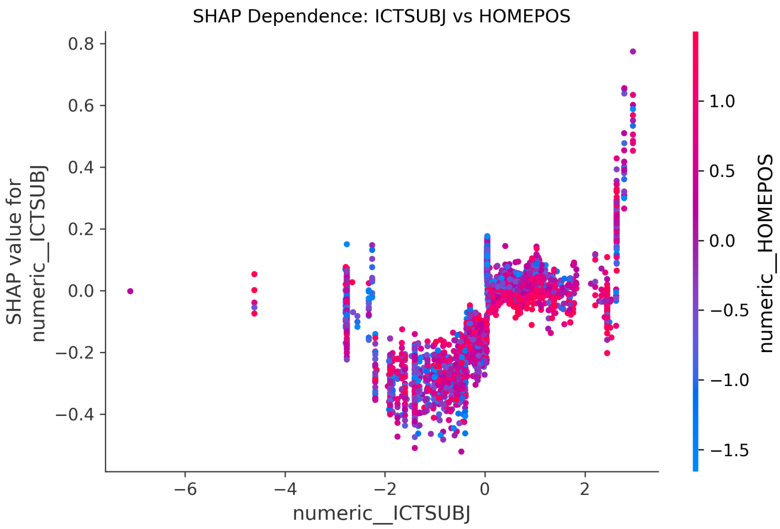

Fig. 13 SHAP dependence: subject-related ICT use (ICTSUBJ) vs. home possessions (HOMEPOS).